Paper Plan

Endogenous Education Choice in Segmented Search Markets

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Core idea. The paper develops and estimates a structural equilibrium model in which workers endogenously choose whether to remain in a low-skill labour market or invest in training that grants access to a higher-skill labour market. Training changes *persistent market eligibility* and interacts with equilibrium search frictions, endogenous match destruction, job-ladder dynamics in the skilled sector, and aggregate regime changes.

The model is the centrepiece. Empirical exercises demonstrate how the framework interprets labour-market dynamics, training decisions, and policy interventions across two qualitatively different crises.

The paper is organised in three parts.

Part I (Sections 1–4) is *conceptual*: it presents the model, describes the data, defines the moments, and discusses identification.

Part II (Sections 5–6) is *methodological*: it explains the multi-regime estimation strategy, defines the parameter classification, and develops the transition-dynamics computation (recovering tightness paths, updating distributions).

Part III (Sections 7–10) is *application*: Application 1 is the financial crisis, Application 2 is COVID-19, followed by a cross-crisis comparison, and Application 3 evaluates education-subsidy policies using the estimated model.

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Part I Conceptual Foundations

1 Introduction

Purpose.

- Motivate the interaction between business cycles and human-capital decisions.
- Emphasise that workers respond to labour-market conditions not only through job search but also through retraining and market switching.
- Highlight that existing search models rarely incorporate endogenous training decisions with persistent market eligibility.
- Argue that the nature of the crisis matters: a demand shock (financial crisis) and a supply/reallocation shock (COVID) affect training incentives through different channels, and a structural model can disentangle them.

Key message. Business cycles influence not only employment outcomes but also the boundary between work and human-capital investment. The channel through which a crisis operates—demand destruction vs. supply disruption—determines whether training incentives rise or fall, with lasting consequences for skill composition.

Contributions.

1. Develop a structural search model with endogenous training and segmented labour markets.
2. Estimate the model on U.S. labour-market data via SMM.
3. Propose a multi-regime estimation strategy—estimate pre-crisis, fix deep parameters, re-estimate regime-specific parameters during crisis—and apply it to two qualitatively different crises.
4. Study transition dynamics after each type of shock.
5. Evaluate education-subsidy policies: compare raising the training flow payoff b_T (stipends) against reducing the cost of education $c(x)$ (investing in higher-education quality).

2 Model

Purpose. Present the economic environment, derive equilibrium conditions, characterise endogenous cutoffs, and describe transition dynamics after regime changes. (The full model is in the companion model notes.)

Key mechanisms.

- Training grants persistent access to the skilled labour market.
- Workers decide whether to train when unemployed in the unskilled sector.
- Skilled workers face job-ladder dynamics through on-the-job search.
- Match destruction occurs endogenously through productivity shocks.
- Aggregate regime changes shift productivity and shock distributions.

Important model outputs.

- Training cutoff $\bar{x}(z)$
- Separation thresholds $p^*(x, z)$ and $p_S^*(x, z)$
- Market tightness $\theta_U(z)$ and $\theta_S(z)$
- Stationary cross-sectional distributions
- Worker transition dynamics after a regime switch

3 Data

Purpose. Describe the data sources used to construct empirical moments. Full variable-level details are in the companion document *Data and Moments*.

Sources.

- Current Population Survey — Basic Monthly (CPS): unemployment rates, skilled share, training share, transition rates (UE, EU, EE).
- CPS Annual Social and Economic Supplement (ASEC): wage moments.
- Job Openings and Labor Turnover Survey (JOLTS): vacancy counts for market tightness.

Estimation windows. Moments are computed over four distinct periods, each approximating a stationary regime:

Window	Period	Role
Pre-FC baseline	Jan 2003 – Dec 2007	Full estimation
Financial crisis	Jan 2008 – Jun 2009	Crisis re-estimation
Pre-COVID baseline	Jan 2015 – Dec 2019	Stability check + 2nd baseline
COVID crisis	Mar 2020 – Dec 2021	Crisis re-estimation

4 Moments and identification

Purpose. Define the 25 empirical moments, explain what each moment identifies in the model, and describe the SMM weight matrix.

4.1 Moment list

The 25 moments are grouped into four blocks:

Labour-market stocks (9 moments). Overall, unskilled, and skilled unemployment rates; skilled share; training share; wage variance and third central moment by skill group. These pin down the stationary distributions and the training cutoff.

Transition rates (6 moments). Job-finding rates (UE) by skill; EU separation rates by skill; skilled EE rate; training rate. The UE rates identify matching efficiencies and tightness. The EU rates identify shock arrival and match-destruction hazards. The EE rate identifies the job-ladder intensity and helps separate the offer distribution Γ_z from p_S^*.

Wages (7 moments). Mean, median, and standard deviation of real hourly wages by skill; log skill premium. Wage levels identify productivity shifters P_U, P_S. The premium identifies the return to training. Dispersion moments identify match-quality distributions and bargaining weights.

Tightness (3 moments). θ_U, θ_S, and θ_S/θ_U. These directly constrain vacancy costs k_U, k_S through free entry.

4.2 Identification intuition

- **Deep parameters** are identified from moments that depend primarily on structural features (wage premium → type distribution; matching elasticities → shape of the JFR–tightness relationship; bargaining weights → wage dispersion).
- **Regime-specific parameters** are identified from moments that shift across windows (unemployment rates, tightness, separation rates, JFR levels).
- The *training share* and *training rate* pin down the training cutoff $\bar{x}(z)$ and the cost function $c(x)$.
- *Higher-order wage moments* (variance, skewness) identify the shape of match-quality distributions.

4.3 Weight matrix

The SMM objective is $Q(\theta) = [\hat{m} - m(\theta)]' \hat{\Xi} [\hat{m} - m(\theta)]$. Initial pass: diagonal $\hat{\Xi} = \mathrm{diag}(1/\hat{\sigma}_1^2, \ldots, 1/\hat{\sigma}_K^2)$ using influence-function standard errors. Second pass: full optimal $\hat{\Sigma}^{-1}$ if well-conditioned.

Part II Estimation and Transition-Dynamics Methodology

5 Multi-regime estimation strategy

This section explains the estimation exercises *before* executing them.

5.1 Parameter classification

Parameters fall into two groups:

Deep structural parameters (constant across all regimes).

- Worker type distribution: a_ℓ, b_ℓ
- Training cost coefficient: a_c
- Matching elasticities: η_U, η_S
- Bargaining weights: β_U, β_S
- OJS flow cost: σ_S

These govern the *technology* and *institutions* of the labour market. They should be invariant to the business cycle.

Regime-specific parameters (free within each window).

- Sector productivity shifters: $P_U(z), P_S(z)$
- Matching efficiencies: $\mu_U(z), \mu_S(z)$
- Vacancy costs: $k_U(z), k_S(z)$
- Shock arrival rates: $\lambda_U(z), \lambda_S(z)$
- Damage/offer shape: $\alpha_U(z), a_\Gamma(z), b_\Gamma(z)$
- Exogenous skilled separation: $\xi_S(z)$
- UI flow values: $b_U(z), b_T(z), b_S(z)$

These capture the *aggregate state* of the economy.

5.2 Stage 1: baseline estimation

Estimate *all* parameters (deep + regime-specific) by matching the model's stationary equilibrium to pre-crisis moments. The baseline window is the pre-FC period (2003–2007).

5.3 Stage 2: crisis re-estimation

Hold the deep parameters fixed at their Stage 1 values. Re-estimate only the regime-specific parameters by matching crisis-period moments. This is done separately for the financial-crisis window (2008–09) and the COVID window (2020–21).

5.4 Structural stability check

Test whether the deep parameters from Stage 1 (estimated on 2003–07) can rationalise the pre-COVID expansion (2015–19) with only regime-specific parameters re-estimated. This is a powerful out-of-sample validation.

6 Transition dynamics: methodology

This section develops the methodology for computing transition paths after an unexpected regime change *in general*, before applying it to any specific crisis.

6.1 The problem

Consider an unexpected, permanent switch from z_0 to z_1 at calendar time T. At T^+, regime-dependent primitives jump to their z_1 values, but the cross-sectional distributions are predetermined (inherited from the old steady state). Equilibrium objects such as $\theta_U(t)$ and the training cutoff $\bar{x}(t)$ must be recomputed using the *current transitional distributions*. The transition is therefore a *joint forward equilibrium path*.

6.2 Steady-state comparison vs. transitional path

Two distinct objects:

(i) Stationary-regime cutoffs. $\bar{x}^{ss}(z_0)$ vs. $\bar{x}^{ss}(z_1)$: how the long-run training rule differs across regimes.

(ii) Transitional cutoff. $\bar{x}(t)$ evolves jointly with tightness and distributions: $\bar{x}^{ss}(z_0) \to \bar{x}(T^+) \to \bar{x}(t) \to \bar{x}^{ss}(z_1)$. The path may be non-monotone.

6.3 Recovering tightness along the transition

At each date t, market tightness is pinned down by free entry given the current composition of job seekers:

Unskilled market.

$$q_U(\theta_U(t)) = \frac{k_U(z_1)}{\int_0^1 J_U(x,1,t)\, u_U(x,t)\, \mathrm{d}x}.$$

Skilled market.

$$q_S(\theta_S(t)) = \frac{k_S(z_1)}{\mathcal{J}_S(t)},$$

where $\mathcal{J}_S(t)$ is the expected value of a contact with a random job seeker drawn from the current pool of unemployed and on-the-job searchers.

Inversion. $\theta_j(t) = (\mu_j / q_j(t))^{1/\eta_j}$. Done at each time step.

6.4 Updating distributions

Given current tightness and policy rules, advance the distributions one step forward:

$$\begin{aligned}
\partial_t\, t(x,t) &= \tau(x,t)\, u_U(x,t) - (\phi+\nu)\, t(x,t),\\
\partial_t\, u_U(x,t) &= \nu\, \ell(x) + \delta_U(x,t)\, e_U(x,t) - \big(f_U(t) + \tau(x,t) + \nu\big)\, u_U(x,t),\\
\partial_t\, u_S(x,t) &= \phi\, t(x,t) + \big(\xi_S + \delta_S^{\text{end}}(x,t)\big)\, e_S^{\text{tot}}(x,t) - \Big(\nu + f_S(t)\big(1 - \Gamma_{z_1}(p_S^*(x,t))\big)\Big)\, u_S(x,t).
\end{aligned}$$

6.5 The transition map

$$(u_U, t, u_S, e_S)_t \implies (\theta_U, \theta_S)_t \implies (U_U^{\text{search}}, U_U, T, U_S, \ldots)_t \implies (\tau, p_U^*, p_S^*, p_S^{oj})_t \implies \partial_t(u_U, t, u_S, e_S).$$

6.6 Implementation

Forward simulation with within-date equilibrium updating:

1. Start from the stationary distribution of regime z_0.
2. Switch primitives to z_1.
3. At each date, solve within-date equilibrium given current distributions.
4. Advance distributions by Δt.
5. Repeat until convergence to the stationary equilibrium of z_1.

Part III Applications

7 Application 1: the financial crisis

7.1 Baseline estimation (2003–2007)

Execute Stage 1: estimate all parameters on pre-FC moments. Report parameter estimates, model fit, and moment-by-moment comparison.

7.2 Crisis estimation (2008–2009)

Execute Stage 2a: fix deep parameters, re-estimate regime-specific parameters on financial-crisis moments.

Expected findings.

- P_U, P_S fall (aggregate productivity decline).
- μ_U, μ_S fall (matching efficiency drops amid uncertainty).
- k_U, k_S may rise (credit tightening raises vacancy-posting costs).
- λ_U rises (more frequent match destruction).
- b_U roughly stable (standard UI was not dramatically expanded).

7.3 Training cutoff: financial crisis

Compute $\bar{x}^{ss}(z_{\text{FC}})$ and compare to $\bar{x}^{ss}(z_0)$.

Hypothesis. The training cutoff *falls* (fewer workers train): both skilled and unskilled prospects worsen roughly symmetrically in a demand shock, and the option value of waiting is high.

7.4 Transition dynamics: financial crisis

Simulate the transition path from the pre-FC steady state to the financial-crisis regime. Plot the time paths of:

- Unemployment rates $\mathcal{U}_U(t)$,$\mathcal{U}_S(t)$
- Training stock $\mathcal{T}(t)$
- Market tightness $\theta_U(t), \theta_S(t)$
- Training cutoff $\bar{x}(t)$
- Skilled share
- Mean wages by sector

8 Application 2: COVID-19

8.1 Pre-COVID baseline (2015–2019)

Execute structural stability check: fix deep parameters from the FC baseline, re-estimate regime-specific parameters on pre-COVID moments. Report fit.

8.2 Crisis estimation (2020–2021)

Execute Stage 2b: fix deep parameters, re-estimate regime-specific parameters on COVID-crisis moments.

Expected findings.

- P_U falls sharply (contact-intensive unskilled sector hit hardest).
- P_S falls less (remote-work compatibility).
- ξ_S spikes (mass temporary layoffs).
- b_U, b_T rise substantially (FPUC, PUA expanded UI benefits).
- λ_U rises sharply (massive job destruction in services).

8.3 Training cutoff: COVID

Hypothesis. The training cutoff *rises* (more workers train): the unskilled sector is disproportionately damaged, expanded UI lowers the cost of the training spell, making the skilled sector relatively more attractive. This provides a structural explanation for the observed enrolment spike.

8.4 Transition dynamics: COVID

Same objects as Application 1, starting from the pre-COVID steady state. Expect a sharp initial spike in unskilled unemployment with partial reversal as temporary layoffs are recalled.

9 Comparing the two crises

With both crises estimated, compare them systematically.

9.1 Parameter shifts

Side-by-side table of regime-specific parameter changes. Which parameters move in the same direction? Which move differently? Is the financial crisis primarily a "vacancy-cost shock" while COVID is a "separation-rate + UI shock"?

9.2 Training-cutoff comparison

$\bar{x}^{ss}(z_{\text{FC}})$ vs. $\bar{x}^{ss}(z_{\text{COVID}})$: opposite-sign predictions? Decompose contributions of each parameter shift to the change in the cutoff.

9.3 Transition-dynamics comparison

Overlay the two transition paths. Compare:

- Speed of adjustment.
- Overshooting / undershooting patterns.
- Which crisis generates longer-lasting hysteresis in the skilled share?

10 Application 3: education-subsidy policies

This is a self-contained counterfactual exercise, separate from the regime-change analysis. We take one of the estimated baselines (pre-FC or pre-COVID) and ask: how does the *stationary equilibrium* respond to two alternative education policies?

10.1 Motivation

The government wants to encourage training. Two policy levers are available, both targeting the training margin but through different channels:

Policy A: higher training flow payoff b_T. The government provides monthly stipends to workers currently enrolled in training (e.g. living-cost grants, tuition waivers that effectively raise disposable income during the training spell). In the model, this is a direct increase in the flow value $b_T(z)$ while the worker is in the training state. It makes training more attractive by reducing the *flow cost* of the training spell (the foregone consumption during the period of non-employment).

Policy B: lower training cost $c(x)$. The government invests in higher-education quality—better equipment, better instructors, improved facilities—at no additional cost to students. For a worker at ability x, the experience of training is now marginally better: the subjective disutility of studying is lower at every ability level. In the model, this is captured as a proportional reduction in the one-time cost function $c(x) = a_c e^{-x}$, i.e. replacing a_c with $(1-\delta)\, a_c$ for some subsidy rate $\delta \in (0, 1)$.

10.2 Key distinction between the two policies

- Policy A operates on the *flow value during training.* It makes the training state less painful on a per-period basis. It is temporary in the sense that the transfer stops once training completes.
- Policy B operates on the *one-time entry cost.* It lowers the up-front hurdle to initiating training. It affects the *threshold* at which a worker is indifferent between searching and training.

Both policies shift the training cutoff $\bar{x}(z)$—more workers will choose to train—but they do so through different channels. Policy A raises the value of the training state $T(x, z)$ by increasing b_T; Policy B lowers the cost $c(x)$ at every x, making the training option $U_U^{\text{train}}(x, z) = -c(x) + T(x, z)$ more attractive.

10.3 Exercise design

1. Fix all parameters at one of the estimated baselines.
2. For a range of subsidy intensities (e.g. 10%, 25%, 50% increase in b_T for Policy A; 10%, 25%, 50% reduction in a_c for Policy B), solve the new stationary equilibrium.
3. Compare outcomes across policies and across intensities.

10.4 Outcomes to report

For each policy and intensity level, report the new stationary-equilibrium values of:

- Training cutoff $\bar{x}$: how far does the cutoff shift? Which policy moves it more per dollar of government expenditure?
- Training stock $\mathcal{T}$: how many additional workers are in training?
- Skilled share $N_S/(N_U + N_S)$: long-run change in skill composition.
- Unemployment rates ur_U, ur_S, and ur_{total}: does encouraging training reduce overall unemployment, or does it create congestion in the skilled market?
- Market tightness θ_U, θ_S: how does vacancy creation respond? Does the unskilled market tighten (fewer searchers) while the skilled market loosens (more entrants)?
- Mean wages by sector: does the inflow of newly trained workers compress the skilled wage premium?
- Separation and job-finding rates: do equilibrium transition rates change?
- Welfare: flow value of workers at various types x.

10.5 Discussion

- Which policy is more effective per unit of fiscal cost? Policy A has a recurring cost (stipends paid every period for every trainee), while Policy B has a one-time capital cost (investing in institutions) that permanently lowers $c(x)$.
- Are there diminishing returns? At high subsidy rates, does one policy saturate faster than the other?
- General-equilibrium feedbacks: does the inflow of trained workers into the skilled market depress skilled wages or tighten the skilled market enough to partially offset the policy?
- Does the ranking of the two policies depend on the current state of the economy (i.e. which baseline we use)? This is a lighter interaction with the regime analysis—not a full regime-change exercise, just a comparison of which baseline produces which ranking.

11 Conclusion

Key takeaways.

- Training decisions respond endogenously to labour-market conditions, and the direction of the response depends on the nature of the shock.
- A demand shock (financial crisis) and a supply/reallocation shock (COVID) operate through different channels, leading to opposite predictions for training take-up.
- Cyclical shocks affect long-run skill composition through hysteresis in the training margin.
- Education subsidies can take the form of flow transfers (stipends) or one-time cost reductions (institutional investment), and the structural model can rank them by effectiveness.
- Structural models that nest regime changes and policy interventions provide a framework for crisis-contingent and investment-oriented labour-market policy.